# B1.1 Carbohydrates and lipids

| NAME | DATE | TOTAL |
|---|---|---|
| | | _____ / 48 |

**About 50 minutes,** or one section at a time. A calculator is allowed. The number of marks shows how many separate points to make.

## Section A: Recall

9 MARKS

1 STATE the maximum number of single bonds that a carbon atom can form. [B1.1.1] [1]

2 NAME the type of reaction that digests polymers. [B1.1.3] [1]

3 STATE the number of carbon atoms in a pentose and in a hexose. [B1.1.4] [2]

4 STATE the monomer of cellulose, and the monomer of glycogen. [B1.1.5, B1.1.6] [2]

5 DEFINE amphipathic. [B1.1.12] [1]

6 STATE the number of water molecules released when a triglyceride forms, and the number released when a phospholipid forms from glycerol and two fatty acids. [B1.1.9] [2]

## Section B: Understanding

20 MARKS

7 EXPLAIN how the properties of glucose suit its use in organisms. [B1.1.4] [4]

8 EXPLAIN how the structure of starch suits it to energy storage in plants. [B1.1.5] [3]

9 **EXPLAIN** how the structure of cellulose suits it to its function. [B1.1.6] [3]

10 **OUTLINE** the role of glycoproteins, with reference to ABO blood groups. [B1.1.7] [3]

11 **EXPLAIN** how the number of double bonds in a fatty acid affects melting point. [B1.1.10] [3]

12 **EXPLAIN** why steroids can pass through a phospholipid bilayer, while ions cannot. [B1.1.13] [2]

13 **DRAW** a section of a phospholipid bilayer, labelling the hydrophilic and hydrophobic regions. [B1.1.12] [2]

## Section C: Working with data

11 MARKS

*All data in this section are invented for practice. The patterns are realistic.*

**14** [B1.1.10]

| FATTY ACID | NUMBER OF CARBONS | C=C BONDS | MELTING POINT / °C |
|---|---|---|---|
| P | 18 | 0 | 70 |
| Q | 18 | 1 | 13 |
| R | 18 | 2 | −5 |
| S | 18 | 3 | −11 |

a **IDENTIFY** which is saturated, and which are polyunsaturated. [2]

b **DESCRIBE** the relationship. [2]

c **SUGGEST** why a fish in cold seas stores mostly fatty acids like R and S. [2]

**15** [B1.1.11]

A person stores about 0.5 kg of glycogen and 12 kg of fat. One gram of glycogen stores 17 kJ of energy, and one gram of fat stores 37 kJ.

a **CALCULATE** the energy stored in each. [2]

b **CALCULATE** the mass of glycogen needed to store the same energy as the fat. [1]

c **EXPLAIN** why the body also keeps a store of glycogen. [2]

## Section D: Longer answer

8 MARKS

16 **COMPARE AND CONTRAST** carbohydrates and lipids as energy stores in organisms. [B1.1.5, B1.1.11] [8]